

Saswat Meher

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Research Interests

Representation learning for information retrieval across different modalities, such as text, images, and graphs.

Education

- 2022–2024 **Indian Institute of Technology, Bombay, India.**
Master of Technology in Computer Science and Engineering
CGPA: 9.69/10
- 2017–2021 **National Institute of Technology, Rourkela, India.**
Bachelor of Technology in Computer Science and Engineering
CGPA: 8.56/10

Work Experience

- 2024–Current **Samsung R&D Institute India, Bangalore, Senior Software Engineer.**
- 2021–2022 **Veritas Technologies LLC, Associate Software Developer.**

Publications

- ICML'25 Indradyumna Roy, **Saswat Meher**, Eeshaan Jain, Soumen Chakrabarti, Abir De (2024). Position: Graph Matching Systems Deserve Better Benchmarks. In Forty-second International Conference on Machine Learning Position Paper Track.
- NeurIPS'24 Eeshaan Jain*, Indradyumna Roy*, **Saswat Meher**, Soumen Chakrabarti, Abir De (2024). Graph Edit Distance with General Costs Using Neural Set Divergence. In The Thirty-eighth Annual Conference on Neural Information Processing Systems.
- LOG'24 Eeshaan Jain*, Indradyumna Roy*, **Saswat Meher**, Soumen Chakrabarti, Abir De (2024). Graph Edit Distance Evaluation Datasets: Pitfalls and Mitigation. In The Third Learning on Graphs Conference.
- Algal Research'20 **Saswat Meher***, Abhijeet Pathy*, Balasubramanian P (2020). Predicting algal biochar yield using eXtreme Gradient Boosting (XGB) algorithm of machine learning methods. Algal Research, 50, 102006.

Research Experience

- Samsung
2024–Current **Gallery Image Retrieval using Natural Language Query.**
Building a Vision-Language Model for gallery image retrieval using natural language queries. Developed a unified X→EN NMT model for 16 languages, replacing separate models and reducing storage by 16× while maintaining comparable accuracy.
- M.Tech.
2024 **Graph Edit Distance for General Cost.**
Advisor: Prof. Abir De, Prof. Soumen Chakrabarti
We propose GRAPHEDX, a novel neural model for approximate GED computation that flexibly incorporates variable edit costs, achieving a 30% reduction in MSE error compared to SOTA.
- M.Tech.
2024 **Knowledge Graph reasoning with Unlimiformer.**
Advisor: Prof. Soumen Chakrabarti
Explored the use of the Transformer-based Unlimiformer model, which claims infinite context length, for Knowledge Graph tasks such as link prediction.

M.Tech. **Compact Subgraph Retrieval for Keyword Query.**

2023 Advisor: Prof. Soumen Chakrabarti, Prof. Abir De

A neural method that uses random walks from terminal nodes to gauge node proximity and iteratively applies a greedy strategy to remove nodes, extracting a compact subgraph from a large graph.

B.Tech **English to Odia Machine Translation using Transfer Learning.**

2020–2021 Advisor: Prof. Ratnakar Dash, NIT Rourkela

Leveraged transfer learning to train a NMT model for Odia-English, a low-resource language.

Course Projects

2024 **Fact Extraction and Verification**, Advisor: Prof. Soumen Chakrabarti.

Built a two-staged retrieval model to verify a given claim by extracting relevant facts from Wikipedia dump.

2023 **Semantic Textual Similarity using BERT and DTW**, Advisor: Prof. Soumen Chakrabarti.

Fine-tuned a BERT-tiny model with Early Interaction for STSB task. Integrated Dynamic Time Warping into the loss to constraint the non-crossing/crossing alignment of tokens.

2023 **Erasing Concepts from Diffusion Model**, Advisor: Prof. Sunita Sarawagi.

Applied Classifier Free Guidance to eliminate a specific concept from a trained Diffusion model. Experimented with Erasing Concepts from DDPM, Conditional DDPM and Stable Diffusion models.

2023 **CLIP with a new logit matching**, Advisor: Prof. Pushpak Bhattacharyya.

Finetuned a CLIP architecture with a different loss and target to handle datasets like Flickr30k with multiple caption for same image for each batch of sample.

2022 **Naive Bayes Document Classifier**, Advisor: Prof. Soumen Chakrabarti.

Built a document classifier using Naive Bayes with Multinomial and Poisson conditional distribution for Cbow and Bbow.

Coursework

Indexing, retrieval and learning for text and graphs(IR), Deep Knowledge Representation, Search, and Complex Inference(Deep IR)

Speech and Natural Language Processing and the Web(NLP), Deep Learning for Natural Language Processing(DLNLP)

Foundations of Machine Learning, Advanced Machine Learning, Learning with Graphs

Honors

2024 **Best TA Award** – CSE Dept., IIT Bombay, for the course 'Deep Knowledge Representation, Search, and Complex Inference.

2017 **Samsung Star Scholarship** – awarded to 300 B.Tech students across India.

Technical Skills

Programming C, C++, Python, Bash

ML Libraries PyTorch, PyG, Tensorflow, NumPy, NetworkX

Responsibilities

Mentoring Student Companion, Institute Student Companion Programme (ISCP), IIT Bombay.

TA **Teaching Assistance at IITB.**

CS347: Operating Systems (Autumn–22)

CS207(M): Discrete Structures (Minor) (Spring-23)

CS635: Indexing, retrieval and learning for text and graphs (Autumn-23)

CS728: Deep Knowledge Representation, Search, and Complex Inference (Spring-24)

Placements Interview Coordinator, Placement Cell, IIT Bombay, Dec 2022